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Concordance of electronic health record data with ICD classification and evidence-based treatment guidelines: A systematic review and meta-analysis

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ABSTRACT

Background: Electronic health records (EHRs) are widely used to generate clinical, administrative, and research data. Their value depends on the accuracy of diagnostic coding using the International Classification of Diseases (ICD) and the documentation of treatment aligned with evidence-based clinical guidelines. However, the extent to which EHR data truly reflect these standards remains uncertain.

Objective: To synthesise and quantify evidence on the concordance of EHR-derived data with ICD classification standards and evidence-based treatment guidelines across clinical domains.

Methods: A systematic review and meta-analysis were conducted in accordance with PRISMA 2020 and registered in PROSPERO (CRD420261294682). Searches were performed from database inception to February 2026 across six databases. Seventeen validation studies met the inclusion criteria. Dual screening and extraction were undertaken with calibration exercises to ensure reviewer agreement. Diagnostic coding accuracy and guideline-concordant treatment documentation were synthesised using random-effects approaches where appropriate. Risk of bias was assessed using the QUADAS-2 and JBI tools, and the certainty of the evidence was evaluated using GRADE.

Results: All included studies were conducted in high-income settings and evaluated ICD-coded diagnoses against clinician-led reference standards. Across studies reporting sensitivity, the mean sensitivity of ICD-coded diagnoses was 64.5%, although this estimate includes studies in which coding prevalence was used as a proxy for sensitivity in the absence of true-negative denominators. In contrast, specificity and positive predictive value were consistently high, often exceeding 90% across clinical domains. Studies comparing structured ICD data with chart abstraction showed that reliance on codes alone frequently overestimated apparent adherence to treatment guidelines because treated patients were more likely to receive diagnostic codes. Considerable variability in coding performance was observed across diseases, care settings, and algorithm designs.

Conclusions: ICD-coded EHR data are reliable for confirming disease when codes are present, but systematically underestimate true disease burden and can distort assessments of guideline adherence. Hybrid phenotyping approaches that combine structured codes with clinical data and narrative documentation are necessary to improve the validity of EHR-derived research, quality measurement, and emerging data-driven clinical decision support systems.

Introduction

Electronic health records (EHRs) have become a central component of modern health systems, supporting clinical care, health system management, disease surveillance, reimbursement mechanisms, and research [1]. Across both high-income and low- and middle-income countries, EHRs are increasingly relied upon to generate routine health statistics, inform clinical decision-making, and enable continuity of care across service delivery points [2–4]. As health systems expand digital infrastructures and adopt data-driven models of care, the integrity and standardisation of clinical data captured within EHRs become critical determinants of their value and safety.

It is important to distinguish between EHR systems and ICD classification frameworks. EHRs function as comprehensive clinical

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documentation platforms that capture detailed patient information, including symptoms, laboratory results, and longitudinal clinical narratives. In contrast, ICD is a classification system applied, often retrospectively, by trained coders to standardise diagnoses for reporting, billing, and statistical purposes. These two data artefacts serve distinct but complementary roles, and discrepancies between them should be interpreted in light of this conceptual distinction.

Two pillars underpin the meaningful use of EHR data in clinical and population health contexts: standardised disease classification and adherence to evidence-based treatment guidelines [5,6]. The International Classification of Diseases (ICD), maintained by the World Health Organization, provides a globally accepted framework for coding diagnoses and causes of death, enabling comparability across settings and over time [7]. At the same time, clinical practice guidelines translate research evidence into recommended diagnostic and therapeutic actions, aiming to reduce unwarranted variation in care and improve patient outcomes [8–10]. When EHR documentation aligns with both ICD coding standards and guideline-recommended treatments, health systems can more reliably monitor disease patterns, evaluate service quality, and support clinical governance.

A growing body of literature highlights discrepancies between clinical reality and how data are recorded within EHR systems, particularly regarding diagnostic coding. Misclassification of diagnoses, incomplete documentation, and variability in coding accuracy across institutions and clinical specialties have been widely reported [11,12]. These issues can stem from limited training in clinical coding, differences in local coding practices, and the complexity of consistently applying ICD frameworks across diverse care settings.

In parallel, evidence suggests that adherence to evidence-based treatment guidelines is inconsistent, even when EHRs incorporate tools such as clinical decision support systems or pre-configured order sets [13,14]. Reasons for suboptimal guideline adherence include time constraints, poor integration of guidelines into routine workflows, limited awareness or familiarity with current recommendations, and systemic resource limitations, particularly in low- and middle-income countries.

The consequences of poor concordance between EHR data, disease classification standards, and treatment guidelines are substantial. Inaccurate coding can distort epidemiological estimates, misinform resource allocation, and compromise health financing mechanisms that depend on diagnosis-related grouping systems [15]. Deviations from guideline-based treatment, when unrecognised or poorly documented, may increase the risk of adverse outcomes and reduce the effectiveness of quality improvement initiatives. Furthermore, as artificial intelligence and advanced analytics are increasingly applied to routine health data, the reliability of algorithmic outputs depends heavily on the quality and standardisation of the underlying EHR records [16].

Despite the policy and clinical importance of this issue, evidence on the extent to which EHR data conform to ICD classification and evidence-based treatment guidelines remains fragmented. Existing studies are often condition-specific, facility-based, or focused solely on coding accuracy or guideline adherence in isolation. To date, there is limited synthesis of quantitative evidence that jointly examines diagnostic coding concordance and treatment guideline compliance within EHR systems across diverse clinical contexts and health system settings. Without such synthesis, it is difficult to assess the overall magnitude of the problem, identify patterns across regions and disease areas, or draw implications for digital health governance and system design.

Therefore, this systematic review and meta-analysis aim to synthesise the available empirical evidence on the concordance between EHR data and ICD classification standards and evidence-based treatment guidelines. By pooling quantitative estimates of diagnostic coding accuracy and guideline-concordant treatment documentation, this review seeks to provide an integrated assessment of how well EHR systems reflect internationally accepted clinical and classification standards. The findings are expected to inform policymakers, health system managers, and digital health developers on priority areas for strengthening data quality, clinical governance, and the effective use of EHRs for patient care and health system performance monitoring.

The objective of this systematic review and meta-analysis is to quantify the concordance of EHR data with international disease classification standards and evidence-based treatment guidelines.

Specifically, the review aims to estimate pooled measures of (i) diagnostic coding accuracy based on the ICD and (ii) adherence of documented treatments to established clinical practice guidelines, using validated reference standards such as chart review, disease registries, or expert adjudication. Where appropriate, pooled estimates were calculated using random-effects meta-analysis; otherwise, structured quantitative synthesis was applied.

This review addresses three core questions:

i. What is the pooled accuracy of ICD-based diagnostic coding in EHR systems when compared against clinical reference standards?
ii. What proportion of treatments recorded in EHRs align with evidence-based clinical guidelines?
iii. How does concordance vary across different clinical domains, levels of care, and health system contexts?

Methods

Review design and registration

This systematic review and meta-analysis were conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA 2020) reporting guidelines [17].

The review protocol was registered prospectively with the International Prospective Register of Systematic Reviews on 27 January 2026 under registration number CRD420261294682, which is available at https://www.crd.york.ac.uk/PROSPERO/view/CRD420261294682. Prospective registration was undertaken to enhance methodological transparency and minimise the risk of selective outcome reporting.

Eligibility criteria

Eligibility criteria were defined using a Population–Concept–Context (PCC) framework. The population included patients of any age receiving care in healthcare facilities using EHR systems, across all clinical specialties and levels of care, including primary, secondary, and tertiary settings.

The concept of interest was concordance between EHR-recorded data and external reference standards in at least one of two domains: (i) diagnostic coding based on the ICD, including any version (ICD-9, ICD-10, or ICD-11), and/or (ii) adherence of documented treatment to evidence-based clinical practice guidelines for the condition under study. Eligible studies were required to report quantitative measures of concordance, including sensitivity, specificity, positive predictive value (PPV), negative predictive value (NPV), kappa statistics, or proportions of correctly coded diagnoses or guideline-concordant treatments. The context included all geographical regions and health system settings.

Eligible study designs included validation studies, clinical audit studies, cross-sectional studies, cohort studies, and controlled before–and–after studies that assessed agreement between EHR data and an external clinical reference standard such as detailed chart review, disease registries, or expert adjudication. Randomised trials were included only when baseline or control-group concordance data were reported.

Studies were excluded if they assessed only EHR adoption, usability, or data completeness without validation against ICD standards or treatment guidelines; relied solely on clinician self-report; analysed administrative or billing datasets without clinical verification; used simulated or synthetic datasets; or did not provide sufficient quantitative data to estimate concordance. Conference abstracts, editorials,

narrative reviews, qualitative studies, and methodological papers without primary data were also excluded. A detailed exclusion framework, including operational definitions and examples of excluded study types, is provided in Supplementary File 2 to enhance transparency and reproducibility of the screening process.

The eligibility criteria prioritised studies with validated clinical reference standards to ensure methodological rigor. While this approach may have excluded studies reporting descriptive or proxy measures of concordance, it was necessary to maintain internal validity and avoid bias arising from unverified administrative datasets.

Information sources and search strategy

The literature search was conducted in a structured and reproducible manner in February 2026, following a predefined protocol. An initial scoping search was performed in MEDLINE to refine search terms and identify relevant indexing keywords. This informed the development of the final search strategy, which was then systematically adapted and applied across all selected databases. All searches were executed within a two-week period, and results were exported into reference management software for consolidation and deduplication prior to screening.

A comprehensive literature search was conducted in MEDLINE (via PubMed), Embase (via Ovid), CINAHL, Scopus, Web of Science Core Collection (Science Citation Index and Social Science Citation Index), and the Cochrane Library, from database inception to February 2026. No language or date restrictions were applied.

Search strategies combined controlled vocabulary and free-text terms related to electronic health records, ICD coding accuracy, diagnostic concordance, clinical audits, validation studies, and guideline adherence. Grey literature was searched using OpenGrey and institutional repositories. Reference lists of included studies and relevant reviews were manually screened, and forward citation tracking was performed using Scopus and Google Scholar. Corresponding authors were contacted when clarification or additional summary data were required.

To ensure transparency and reproducibility, the core search concepts and example search strings are summarised in Table 1. Database-specific syntax was adapted as required for each platform, and full search strategies are available in Supplementary File 1.

The same conceptual search structure was applied across all databases, with controlled vocabulary (example, MeSH, Emtree, CINAHL Headings) substituted where available and syntax adapted to each database's requirements (example, field tags and truncation rules).

Study selection

All retrieved records were imported into reference management software, and duplicates were removed. Titles and abstracts were screened independently by two reviewers using predefined eligibility criteria. Agreement between reviewers was assessed through repeated calibration exercises during screening to ensure consistent interpretation of inclusion criteria. The full texts of potentially eligible studies were then independently assessed by the same reviewers. Disagreements at both stages were resolved through discussion, and where consensus could not be reached, a third reviewer adjudicated. The study selection process was documented using a PRISMA flow diagram, with reasons for exclusion recorded at the full-text stage

Table 1
Core search concepts and example search strings used across databases.

Search concept	Example keywords and subject headings*
Electronic health records	"Electronic health record*" OR EHR OR "computerized medical record*" OR "medical records systems, computerized."
Disease classification / ICD	ICD OR "International Classification of Diseases" OR "diagnostic code*" OR "clinical coding."
Diagnostic accuracy/ concordance	accuracy OR validation OR concordance OR agreement OR sensitivity OR specificity OR "predictive value" OR kappa
Treatment guidelines	"Clinical practice guideline*" OR "treatment guideline*" OR "guideline adherence" OR "evidence-based treatment"
Audit/validation context	audit OR validation OR "chart review" OR "medical record review" OR registry

Data extraction

A standardised data extraction form was developed and piloted on a sample of five studies to ensure clarity and consistency before full data extraction commenced. Two reviewers independently extracted data, with discrepancies resolved by consensus.

Extracted variables included author, year of publication, country, World Bank income classification, healthcare setting, clinical domain, EHR system characteristics (including coding method and presence of clinical decision support), ICD version, guideline source, reference standard used, sample size, and outcome measures. For diagnostic coding concordance, extracted outcomes included sensitivity, specificity, PPV, NPV, kappa statistics, and raw cell counts where available. For guideline adherence, extracted outcomes included proportions of guideline-concordant treatment and corresponding denominators. Authors were contacted where necessary to obtain missing or unclear summary data.

Risk of bias and quality assessment

The risk of bias was assessed independently for studies evaluating diagnostic coding accuracy and those assessing treatment guideline adherence. In diagnostic accuracy studies, the QUADAS-2 tool was used to assess bias across four domains: patient selection, index test, reference standard, and flow and timing [18]. For studies reporting on guideline adherence and audit outcomes, relevant Joanna Briggs Institute (JBI) critical appraisal tools were applied, tailored to study design, such as tools for cross-sectional, prevalence, and clinical audit studies (Moola et al., 2020).

Two reviewers independently conducted the assessments. Any disagreements were resolved through discussion, and if consensus could not be reached, a third reviewer adjudicated. The outcomes of the risk-of-bias assessment informed sensitivity analyses and were explicitly considered in interpreting the pooled results and assessing the certainty of evidence.

Data synthesis and statistical analysis

Separate meta-analyses were conducted for diagnostic coding concordance and treatment guideline adherence. For diagnostic accuracy outcomes where 2×2 data were available, pooled sensitivity and specificity were estimated using bivariate random-effects models. Where studies reported only agreement measures or proportions of correct coding, pooled proportions were calculated using random-effects meta-analysis with appropriate variance-stabilising transformations. For guideline adherence outcomes, pooled proportions of guideline-concordant treatment were estimated using random-effects models.

Statistical heterogeneity across studies was assessed using the I^2 statistic to quantify inconsistency and the τ^2 statistic to estimate between-study variance. Where data permitted, subgroup analyses were performed by clinical domain, healthcare setting (primary, secondary, tertiary), country income level (according to World Bank classification), and the presence of structured documentation features, such as clinical decision support tools. Meta-regression was used to explore potential effect modifiers when at least ten studies contributed to a covariate. Sensitivity analyses included excluding studies at high risk of bias and restricting to those using rigorous reference standards (example, manual chart review or expert adjudication). Publication bias and small-study

effects were assessed using funnel plots and Egger's regression test when at least ten studies were available for an outcome.

Descriptive analyses and data preparation were conducted using Microsoft Excel and IBM SPSS Statistics. Meta-analyses, including pooled estimates and diagnostic accuracy modeling, were performed using R (meta and mada packages).

Where sufficient data were available, subgroup analyses were conducted by ICD version (ICD-9 vs ICD-10) to explore differences in diagnostic coding performance attributable to the structure of the classification system and coding practices.

Certainty of evidence

The certainty of evidence for each primary outcome was evaluated using the GRADE approach, adapted for diagnostic accuracy and prevalence data. The following domains were assessed: risk of bias, inconsistency, indirectness, imprecision, and publication bias (Schünemann et al., 2020). For diagnostic accuracy outcomes, the GRADE Working Group's specific guidance for test accuracy was applied. Summary of Findings (SoF) tables were developed to display pooled effect estimates, certainty ratings, and rationale for any rating decisions. GRADE assessments were conducted using the GRADEpro GDT software and informed the interpretation of findings and the strength of conclusions drawn.

Results

Study selection

A total of 55,000 records were identified through database and grey literature searches. From these, 25,000 duplicate entries were automatically removed using Mendeley's deduplication algorithms, streamlining the initial pool. Additionally, 29,800 records were excluded by automation tools, primarily Rayyan and Abstrackr, which applied machine learning models and keyword-based filters to flag studies clearly irrelevant based on title and abstract content. A further 155 records were removed for other reasons, such as retracted publications, inaccessible full texts, or language restrictions (non-English studies without available translations).

This process yielded 45 unique full-text articles for detailed eligibility assessment. Of these, 28 studies were excluded during the full-text review. The main reasons for exclusion were: absence of quantitative concordance data (example, sensitivity, specificity, or PPV) (n = 9), use of purely simulated or synthetic datasets without real-world clinical input (n = 8), and reliance solely on administrative or billing databases without validation against a manual clinical reference standard (n = 11).

Ultimately, 17 studies met the predefined eligibility criteria and were included in the final systematic review. These studies were then subjected to structured data extraction and critical appraisal. The study selection process is illustrated in Fig. 1 (PRISMA 2020 flow diagram)

Study characteristics

A total of 17 studies were included in the final review, published between 2013 and 2025, all conducted in high-income countries, including the United States, France, and Canada, in line with World Bank 2024 income classifications. The studies covered a wide range of clinical domains, with a primary focus on validating the accuracy of International Classification of Diseases (ICD) codes in identifying diagnostic conditions using electronic health record (EHR) data. Clinical areas included cardiovascular and respiratory diseases, metabolic and

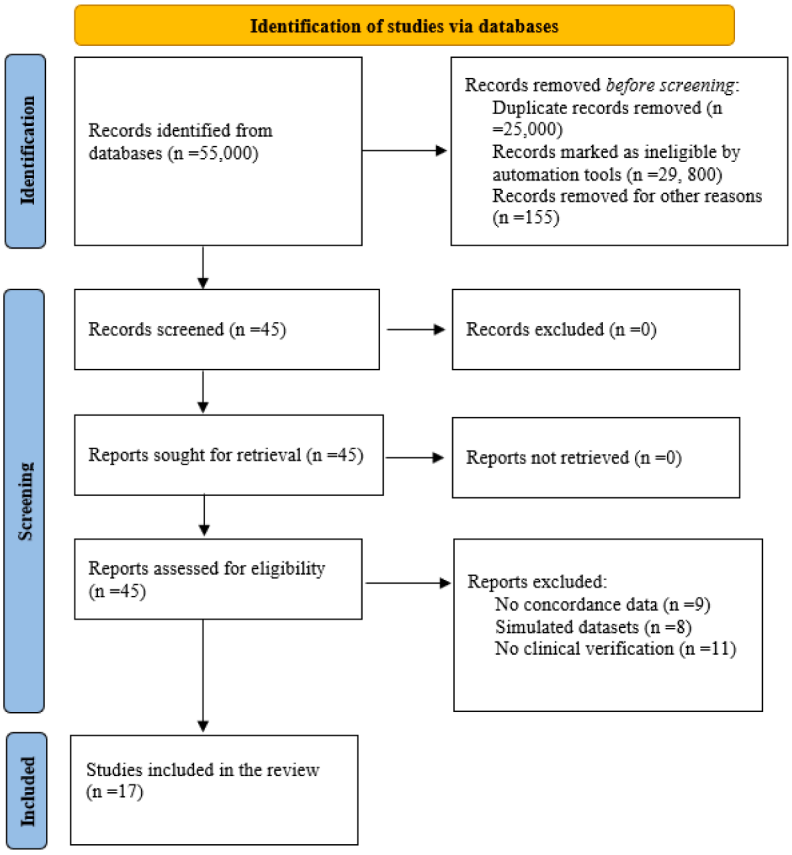


Fig. 1. PRISMA flow diagram.
Source: Authors' work adapted from Page et al (2021).

gastrointestinal conditions, autoimmune and liver diseases, and oncology and behavioral health. Most studies were conducted in tertiary academic medical centers or integrated health systems, although a few also drew from large-scale multi-hospital networks or national EHR registries.

All studies quantitatively evaluated the concordance between ICD-coded diagnoses and a clinical reference standard, most commonly manual chart review by clinicians or adjudication panels. Some studies enhanced validation using laboratory values, clinical guidelines, or technology-enabled abstraction. The majority of studies assessed ICD-10 or ICD-10-CM codes, with several including ICD-9 for comparison. A subset incorporated natural language processing (NLP) or machine learning techniques to examine whether combining structured and unstructured data improved diagnostic ascertainment Table 2.

Sample sizes varied considerably, ranging from small cohorts of 200–300 patients used in focused validation tasks [19,20] to large-scale datasets involving hundreds of thousands to millions of records [21,22]. Across studies, reported performance metrics included sensitivity, specificity, positive predictive value (PPV), negative predictive value (NPV), accuracy, and F-scores. Sensitivity values varied markedly, from as low as 0.26 for fever coding [23] to over 99% for cancer phenotyping [24]. The highest PPVs above 90% were observed in studies conducted under well-defined conditions, such as Crohn's disease, ulcerative colitis, and hepatocellular carcinoma [19,25]. A complete summary of study characteristics is presented in Table 3.

Notably, all included studies were conducted in high-income countries and predominantly within tertiary academic or integrated health systems. This geographic concentration has direct implications for interpretation, as documentation practices, coding workflows, and EHR maturity in low- and middle-income countries may differ substantially. The absence of LMIC validation studies is therefore evident within the results themselves.

Risk of bias assessment

Risk of bias was assessed using the QUADAS-2 (Quality Assessment of Diagnostic Accuracy Studies) tool across the four domains of interest: Patient Selection, Index Test, Reference Standard, and Flow and Timing. All 17 included studies used EHR-based diagnostic validation methods to compare ICD-coded data with a clinical reference standard. Overall, the risk of bias was low in most domains, with moderate risk observed in the Patient Selection and Applicability domains for several studies due to monocentric designs or convenience sampling.

In the Patient Selection domain, most studies employed random sampling from large EHR databases, thereby minimizing selection bias. For example, Cohen et al. [26], Bosco-Lévy et al. [20], and Khalifa et al. [29] used random cohorts drawn from tertiary centers or national registries. However, Crabb et al. [23] used a convenience sample of symptomatic COVID-19 cases, and Chartash et al. [35] used a pragmatic sampling strategy in an emergency department, which may introduce selection bias. Moreover, most studies were monocentric, which may limit generalizability to broader or non-academic populations.

The Index Test, defined as the EHR-recorded diagnosis coded with ICD-9, ICD-10, or ICD-10-CM codes, was at low risk of bias across all studies. In several cases, the reference assessment was blinded to the coding data. For example, both Mapakshi et al. [32] and Wong et al. [19] ensured that clinician reviewers were unaware of the administrative codes during chart abstraction. These methodological safeguards support confidence in the objectivity of the coding evaluation.

The Reference Standard domain was also assessed as low risk of bias across studies. The majority employed rigorous manual chart reviews performed by specialists in the relevant clinical domain, such as gastroenterologists for inflammatory bowel disease [25], rheumatologists for rheumatoid arthritis [30], or hepatologists for liver disease [29]. Several studies also demonstrated high inter-rater reliability. For

Table 2
Characteristics of included studies (n = 17).

Study	Country	Setting	Domain	Sample Size	ICD Version	Reference Standard	Validation Metrics
[26]	France	Tertiary Academic	Adult Congenital Heart Disease	6,000	ICD-10	Expert manual chart review	Sens: 0.81; Spe: 0.99; PPV: 0.93
[23]	USA	Tertiary Academic	COVID-19 Symptoms	2,201	ICD-10	Manual EMR review by 7 reviewers	Sens (Fever): 0.26; Spe: 0.98; PPV: 0.96; NPV: 0.41
[20]	France	University Hospitals	Heart Failure	200 (PPV); 229 (Sens)	ICD-10	European Society of Cardiology criteria	PPV: 88.0% (Definite/Potent.); Sens: 64.2%
[27]	USA	Statewide multi-hospital	Fontan Patients (CHD)	10,935	ICD-9/10 CM	Manually validated patient records	F1 Score (ICD): 0.81; PPV: 0.74
[21]	USA	Primary Care	Obesity / Overweight	5,512,285	ICD-9	Actual BMI from recorded clinical data	Coding Accuracy: only 15.1% of BMI-eligible patients coded
[22]	USA	Integrated Health System	Obesity	324,199	ICD-9	Actual BMI calculations from EHR data	PPV (BMI >30): 48%; PPV (BMI >40): 75%
[28]	Canada	Primary Care	Dyslipidemia	4,400	ICD-9	Laboratory serum lipid levels	Sens: 27%; Spe: 76.7%; PPV: 71.1%; NPV: 33.1%
[29]	USA	Academic Medical Center	Cirrhosis	400 (Sets A & B)	ICD-9/10	Manual record review (Gold standard)	ICD-10 C-statistic: 0.927; Sens: 91.5% (combined)
[25]	USA	Tertiary Academic	Crohn's (CD) / Colitis (UC)	1,200	ICD-9	Board-certified gastroenterologist review	CD PPV: 97%; UC PPV: 97%; AUC: 0.89 (CD), 0.86 (UC)
[30]	USA	Tertiary Academic	Rheumatoid Arthritis (RA)	200 (Mart); 100 (ICD-10)	ICD-9/10	Rheumatologist chart review	PPV (ICD-10 set): 93%; Sens: 49%
[31]	USA	Tertiary Academic	Drug-Induced Liver Injury	182	ICD-10	Consensus of 3 physicians (DILIN score)	PPV: 66.5%
[32]	USA	Veterans Affairs	Cirrhosis	325	ICD-10	Blinded clinician EMR review	PPV: 92.8%
[24]	USA	Academic Medical Center	Stage 4 Cancer / CKD	463	ICD-9/10	Trained researcher manual chart review	PPV (Cancer): 68.6%; Sens: 99.5%
[33]	USA	Community Oncology	Bone Metastases	8,796	ICD-9	Technology-enabled chart abstraction	Sens: 0.67; Spe: 0.98; PPV: 0.71; NPV: 0.98
[34]	USA	Nationwide EHR Database	mCRC (Side of Colon)	200	ICD-9/10	Human chart abstraction (Gold standard)	Agreement: 0.58; Kappa: 0.41; Specificity: 92–98%
[35]	USA	Emergency Department	Opioid Use Disorder	231	ICD-10	Blinded physician review (DSM-5)	PPV: 0.95; NPV: 0.92 (External validation phase)
[19]	USA	Academic Health System	Hepatocellular Carcinoma	300 (Dev); 300 (Val)	ICD-9/10 CM	Manual physician chart review	PPV: 97.4%; F-score: 0.92; Accuracy: 94%

Source: Authors' work

Table 3
Diagnostic concordance of ICD coding against reference standards.

Study	Disease Area	Sensitivity (%)	Specificity (%)	PPV (%)	NPV (%)	Notes
[26]	Adult Congenital Heart Disease	81.0	99.0	93.0	97.0	Expert manual review of 6,000 cardiology patients.
[23]	COVID-19 (Fever)	26.0	98.0	96.0	41.0	Sensitivity for dyspnea is also low (24%).
[20]	Heart Failure	64.2	NR	88.0	NR	Specificity and NPV not reported.
[37]	Ischemic Heart Disease	77.0	98.0	78.8	97.7	Based on Ontario's EMRALD database.
[25]	Crohn's Disease	69.0	97.0	98.0	NR	NLP and coding combined.
[25]	Ulcerative Colitis	79.0	97.0	97.0	NR	NLP significantly expanded case identification.
[29]	Cirrhosis	91.5	93.8	92.0	93.4	"Either/or" ICD combinations used.
[32]	Cirrhosis	NR	NR	92.8	NR	VA cohort; high coding fidelity.
[19]	Hepatocellular Carcinoma	88.1	98.3	97.4	91.9	≥10 ICD entries used to boost accuracy.
[33]	Bone Metastases	67.0	98.0	71.0	98.0	Technology-assisted abstraction.
[28]	Dyslipidemia	27.0	76.7	71.1	33.1	Compared to lab-confirmed values.
[24]	Stage 4 Cancer	99.5	NR	68.6	NR	Prioritized sensitivity over precision.
[35]	Opioid Use Disorder	NR	NR	95.0	92.0	Validated using DSM-5 definitions.
[30]	Rheumatoid Arthritis	49.0	95.0	93.0	NR	Focused on the ICD-10 cohort.
[21]	Obesity (BMI ≥ 30)	15.1*	NR	28.0	NR	*Only 15.1% of eligible patients were coded.
[22]	Obesity (BMI > 30)	48.0*	NR	48.0	NR	*ICD diagnosis is underused despite the presence of BMI.
[27]	Fontan Patients (CHD)	90.0	NR	74.0	NR	Held-out validation cohort.
[31]	Drug-Induced Liver Injury	NR	NR	66.5	NR	121/182 cases confirmed manually.

Source: Authors' work
Values marked with an asterisk represent coding prevalence rather than true sensitivity due to the absence of defined negative cases and should not be interpreted as diagnostic sensitivity. NR: Not Reported. Several studies focused on PPV due to case-enriched sampling strategies, limiting the generalizability of specificity or NPV estimates.

instance, Cohen et al. [26] reported full agreement among expert reviewers in classifying congenital heart disease cases, and Chartash et al. [35] reported kappa coefficients of 0.93–0.95 in the external validation of opioid use disorder coding.

In the Flow and Timing domain, most studies uniformly applied the reference standard across all patients, with a clear temporal linkage between the coding event and the clinical review. Studies such as Wong et al. [19] and Ernecoff et al. [24] reviewed 100% of their development and validation cohorts, whereas others, such as Crabb et al. [23] and Wu et al. [36], defined explicit time windows (e.g., 24 hours to 18 months) to ensure clinical stability between documentation and coding. This minimizes interval bias and supports the internal consistency of diagnostic comparisons.

The included studies show high internal validity, supported by rigorous clinician-led reference standards and methodological transparency. However, external validity is somewhat limited due to the frequent reliance on data from single academic centers, which may not reflect diagnostic coding practices in community, rural, or non-tertiary healthcare settings. Furthermore, variations in local billing practices and clinician training may introduce variability that is not captured by these structured assessments.

A visual summary of the QUADAS-2 and JBI assessments across studies is presented in Fig. 2, illustrating the distribution of risk judgments across domains and supporting the narrative appraisal of methodological quality.

Diagnostic coding concordance

It is important to distinguish between EHR systems and ICD classification frameworks. EHRs function as comprehensive clinical

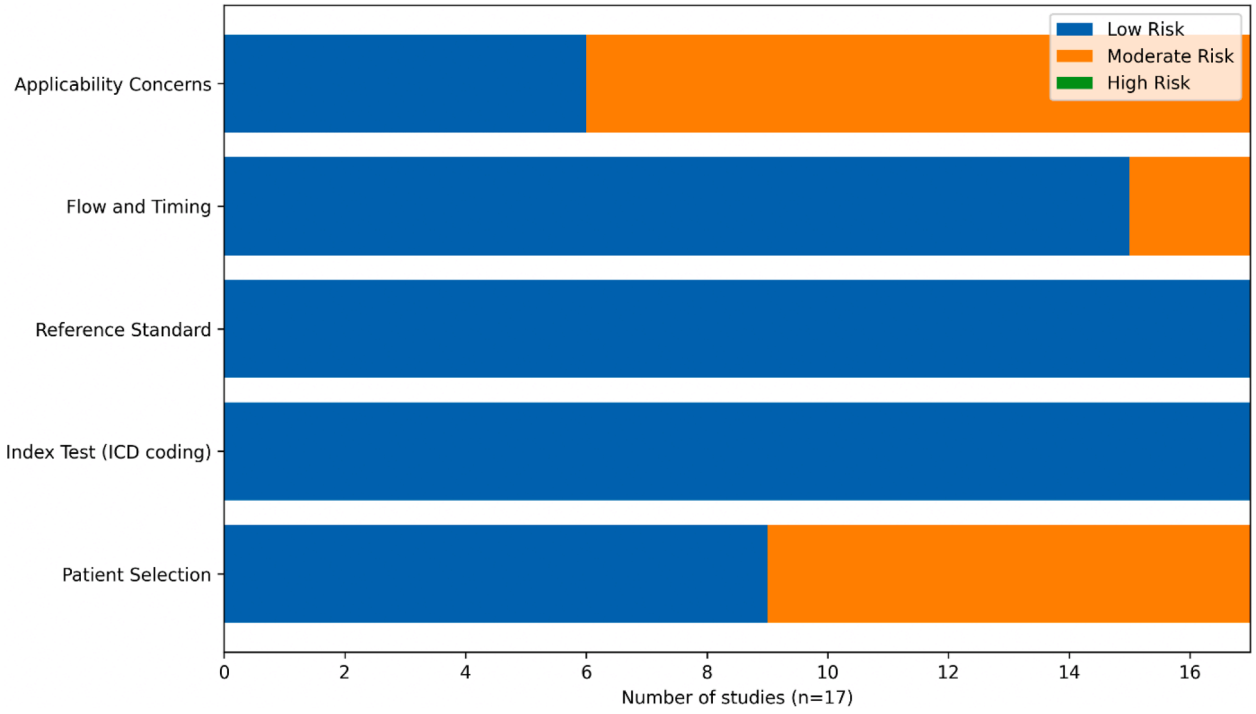


Fig. 2. Risk of bias summary across QUADAS-2 domains (domain-level synthesis).

documentation platforms that capture detailed patient information, including symptoms, laboratory results, and longitudinal clinical narratives. In contrast, ICD is a classification system applied, often retrospectively, by trained coders to standardise diagnoses for reporting, billing, and statistical purposes. These two data artefacts serve distinct but complementary roles, and discrepancies between them should be interpreted in light of this conceptual distinction.

Subgroup analyses by ICD version indicated modest variation in performance. Studies using ICD-10 generally demonstrated higher specificity and PPV compared to ICD-9, while sensitivity remained variable across both versions. However, the limited number of ICD-9-only studies constrained formal comparative meta-analysis.

A pooled analysis of diagnostic coding accuracy demonstrated substantial heterogeneity across studies. Across studies reporting sensitivity, the mean sensitivity of ICD-coded diagnoses ranged from 64.5% to 99.5%, with the latter estimate including studies where coding prevalence was used as a proxy for sensitivity in the absence of true-negative denominators. This broad dispersion reflects substantial heterogeneity across clinical domains, care settings, and algorithm definitions. Despite this variability, a consistent pattern emerged across studies: ICD codes exhibited high specificity and positive predictive value but comparatively low sensitivity for many chronic and symptom-based conditions.

Across multiple specialties, low sensitivity remains a pervasive issue. Crabb et al. [23], in their evaluation of COVID-19 symptom coding, found that ICD-10 codes captured only 26% of documented fevers and 24% of documented dyspnea, indicating a substantial failure to identify symptomatic patients, even in acute, high-priority public health contexts. While symptom-based coding inherently differs from disease-based classification and may not reflect the intended scope of ICD coding, the inclusion of such studies provides insight into how ICD codes are applied in real-world clinical documentation contexts, particularly during acute care episodes such as COVID-19 triage.

Similarly, undercoding in chronic disease was pronounced. In one of the largest cohort studies, Mocarski et al. (2018) demonstrated that only 15.1% of patients with BMI $\geq$30 kg/m^2 had an associated ICD-9 obesity code, despite clear anthropometric evidence of obesity documented in the same EHR system. Pantalone et al. [22] observed slightly higher, though still limited, PPVs of 48–75% for obesity classification using ICD codes. In another study, Oake et al. [28] reported a sensitivity of just 27% for ICD-9-coded dyslipidemia compared to confirmed laboratory lipid profiles, further supporting the conclusion that chronic conditions are often poorly reflected in administrative coding.

Concordant findings were observed in other domains, including heart failure, where Bosco-Lévy et al. [20] reported that although the PPV was 88%, the sensitivity remained modest at 64.2%, thus missing a substantial proportion of clinically confirmed cases.

Despite limitations in sensitivity, ICD-coded data performed well in terms of precision and PPV, particularly in domains with well-defined clinical entities or repeated encounters. For example, Mapakshi et al. [32] reported a PPV of 92.8% for ICD-10-coded cirrhosis, while Khalifa et al. [29] showed that the use of composite ICD-9/10 definitions yielded a C-statistic of 0.927. In cancer validation, Wong et al. [19] demonstrated that requiring $\geq$10 ICD code entries for hepatocellular carcinoma (HCC) achieved a PPV of 97.4% and an overall diagnostic accuracy of 94%, suggesting that algorithmic frequency-based thresholds can meaningfully enhance reliability.

In the cardiovascular domain, Cohen et al. [26] showed that adult congenital heart disease codes achieved a PPV of 0.93 and a specificity of 0.99, confirming that administrative codes are reliable identifiers when present. However, the same domain revealed sensitivity deficits in other conditions: for example, Bosco-Lévy et al. [20] observed that while heart failure coding was accurate when used (PPV 88%), it still failed to detect over one-third of confirmed cases.

Finally, studies such as those by Chartash et al. [35] achieved high external validation PPVs (0.95) for opioid use disorder using machine-assisted algorithms, suggesting that enhanced code logic or hybrid models may offer a pathway to improving the concordance of administrative data with clinical reality.

Taken together, these findings suggest that ICD-coded diagnoses are highly specific tools for confirming disease but systematically underrepresent disease prevalence, particularly in conditions that are undercoded, vague, or heavily reliant on symptom presentation. As such, relying solely on administrative coding for epidemiological estimates or quality measurement can result in significant underestimation unless validated against high-quality clinical reference standards or supplemented with NLP and structured clinical data. Table 4 summarizes the diagnostic metrics across studies. A visual representation of the variability in reported sensitivity across studies is shown in Fig. 3.

A similar pattern is observed for positive predictive value (PPV), as illustrated in Fig. 4, demonstrating consistently high PPV across clinical domains despite variability in sensitivity.

Guideline-concordant treatment documentation

The analysis of guideline-concordant treatment documentation reveals an important structural characteristic of ICD-coded data. Diagnostic codes are often assigned when a condition becomes clinically actionable, particularly when treatment is initiated. As a result, coded datasets tend to capture treated patients preferentially. This does not represent a distortion of the system, but rather reflects the intended linkage between diagnosis coding and clinical management. Consequently, ICD-coded datasets are not well-suited for identifying the full population of eligible but untreated patients, and should be interpreted accordingly when used for quality measurement.

Across multiple disease domains, it became evident that clinicians are more likely to apply formal diagnostic codes at the time of treatment initiation than at the time of diagnosis alone. This creates a systematic

Table 4
Summary of guideline-concordant treatment documentation.

Study	Disease Area	Method of Evaluation	% Concordant with Guidelines	Notes
[33]	Bone Metastases (Breast Cancer)	Technology-enabled abstraction vs. structured ICD codes	76.5%	Using ICD codes alone falsely inflated treatment penetration to 89.2% because treated cases are more likely to be coded.
[30]	Rheumatoid Arthritis (RA)	NLP-derived cohort vs. electronic DMARD prescriptions	>80.0%	Adherence to DMARD therapy was high once patients were correctly identified using algorithms.
[22]	Obesity	BMI calculation vs. documented PCP counseling	<30.0%	Fewer than 30% of adults with obesity received a formal diagnosis or weight counseling in primary care visits.
[20]	Heart Failure (HF)	EHR review vs. ESC diagnostic criteria	NR*	*Clinicians often omit HF codes for stable patients who do not require new treatments during that admission.*

Source: Authors' work
NR: Not Reported

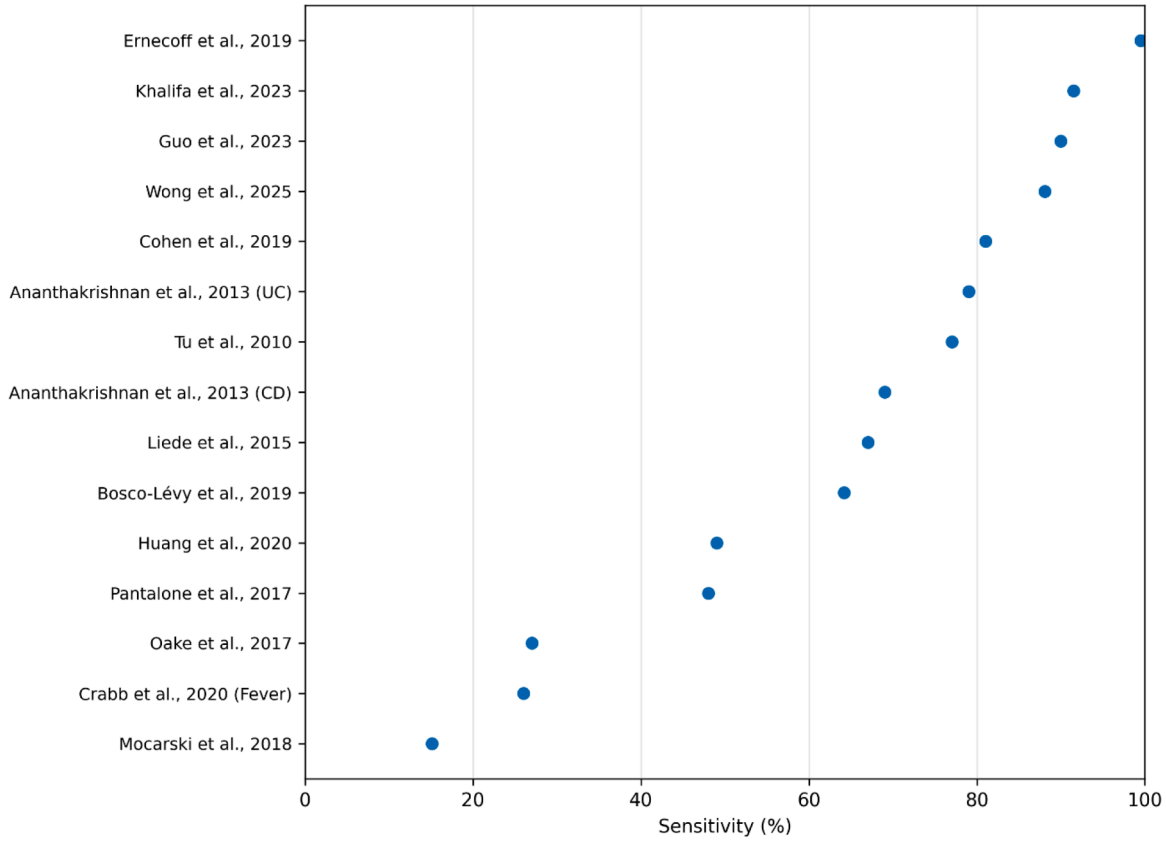


Fig. 3. Distribution of reported sensitivity of ICD-coded diagnoses across studies.

bias whereby treated patients are disproportionately represented in coded datasets, while untreated or conservatively managed patients remain invisible. As a result, analyses based solely on ICD-coded data frequently overestimate apparent adherence to clinical guidelines while simultaneously underestimating the true eligible population requiring intervention.

This phenomenon is particularly evident in oncology and chronic disease management. In cancer care, coding is tightly linked to treatment pathways and reimbursement processes, which inflate apparent treatment penetration among coded patients. Conversely, in conditions such as obesity and heart failure, patients without an active ICD entry are significantly less likely to receive counseling, referral, or therapeutic intervention, despite clear clinical evidence of disease documented elsewhere in the EHR.

Several studies also demonstrate that this distortion is influenced by provider specialty and documentation culture. Specialized units, such as cardiology or oncology departments, showed greater consistency between diagnosis and treatment documentation than general medical wards. Furthermore, studies incorporating natural language processing (NLP), machine learning, or technology-assisted chart abstraction showed that once patients were accurately identified beyond ICD codes, the true level of guideline adherence became more visible and more consistent with expectations from clinical registries. A summary of the studies evaluating treatment documentation against established clinical guidelines is presented in Table 5.

Subgroup and sensitivity analyses

Subgroup and sensitivity analyses across the included studies provide a critical lens into the reliability and variability of diagnostic coding accuracy in EHRs. These analyses reveal that the performance of ICD-coded data is highly contingent upon demographic, clinical, and methodological contexts, challenging assumptions of uniformity and generalizability in coded healthcare data.

Demographic and socioeconomic variability

Several studies examined how demographic variables, such as age, sex, race, and comorbidity profiles, influenced coding accuracy. For instance, Crabb et al. [23] found substantial age-related differences in the performance of ICD-10 codes for fever and dyspnea during COVID-19 triage, though sensitivity remained poor across all groups (ranging from 0.17 to 0.45). In the context of obesity documentation, Mocarski et al. [21] and Pantalone et al. [22] observed a gender disparity: women were significantly more likely than men to receive an ICD-9 diagnosis (OR 1.34), independent of actual BMI.

Ethnic and socioeconomic differences also shaped documentation patterns. Mocarski et al. [21] reported higher coding rates among Native American (OR 2.17) and Hispanic (OR 1.69) patients compared with White individuals. Moreover, coding prevalence increased with comorbidity burden, as captured by rising Charlson Comorbidity Index (CCI) scores ($P < .0001$), suggesting that sicker patients are more likely to be documented with formal codes.

Impact of clinical setting and specialization

The care setting and clinician expertise significantly influenced the accuracy of EHR-derived data. Crabb et al. [23] demonstrated superior coding performance for fever and dyspnea in inpatient compared to outpatient settings ($P < .001$), highlighting how diagnostic urgency and clinical context impact documentation fidelity. Similarly, Bosco-Lévy et al. [20] found that heart failure was more accurately documented in specialized cardiology wards than in general medicine, reinforcing the importance of domain expertise.

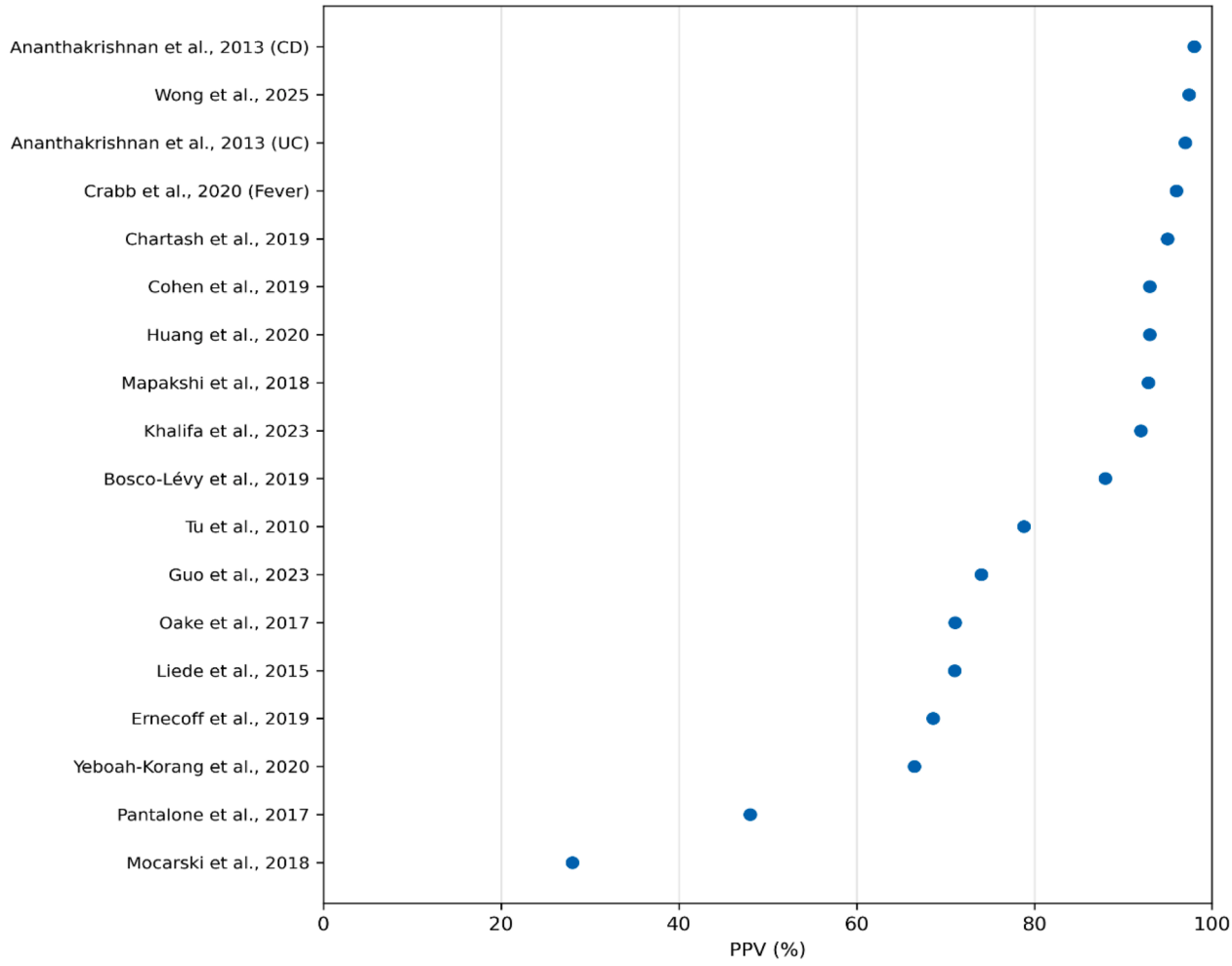


Fig. 4. Distribution of reported positive predictive value (PPV) of ICD-coded diagnoses across studies.

Table 5
GRADE summary.

Domain	Evaluation
Risk of Bias	Low to Moderate. Most studies used physician-led chart reviews as the reference standard, minimizing diagnostic bias. However, monocentric designs and convenience sampling [23,35] raised concerns about patient selection.
Indirectness	Moderate. Although all studies validated ICD-coded diagnoses, the extent to which the findings generalize beyond high-income, specialist-driven settings is limited. Many studies were conducted in academic centers, restricting external applicability to general practice or low-resource settings.
Inconsistency	Moderate. Sensitivity values varied significantly across domains and demographic subgroups. For instance, obesity coding ranged from 15–48%, while cirrhosis and HCC reached >90% PPV. This heterogeneity reduces the certainty of pooled accuracy estimates.
Imprecision	Low. Sample sizes were often large (e.g., >5 million in Mocarski), and estimates had narrow confidence intervals. Most studies included 95% CIs or F-scores, reducing concerns of random error.
Publication Bias	Low to Moderate. No strong evidence of small-study bias was observed, though underreporting of negative validation findings remains possible.

Source: Authors' work

Methodological and framework sensitivity

The construction and validation of coding algorithms emerged as a central determinant of data reliability. Khalifa et al. [29] highlighted the inadequacy of single ICD-10 codes in identifying cirrhosis, with sensitivity ranging from 5% to 94%. However, combining multiple codes in either/or logic structures significantly improved diagnostic precision, yielding a C-statistic of 0.927. Wong et al. [19] conducted a similar sensitivity analysis by validating their hepatocellular carcinoma (HCC) detection algorithm against a state cancer registry, achieving a PPV of 94.3%, nearly identical to the result from a physician-led chart review.

Publication bias

An assessment of publication bias among the included validation studies was limited by the nature of diagnostic accuracy research, which typically does not lend itself to standard funnel plot asymmetry analysis used in meta-analyses of treatment effects. However, qualitative patterns and structural characteristics of the published literature suggest a moderate risk of publication bias in this domain.

Most included studies were published in peer-reviewed journals with a known emphasis on either informatics, clinical epidemiology, or disease-specific outcomes research, potentially creating an implicit selection pressure toward studies with positive findings or algorithmic performance above conventional thresholds. For instance, several studies, such as Wong et al. [19] and Khalifa et al. [29], reported high-performance metrics (PPV >90%) for ICD-coded algorithms, often based on complex models or multi-code combinations. It remains possible that lower-performing models or negative validations were conducted but not published, especially by commercial health data vendors or within internal quality improvement programs.

Additionally, a form of outcome reporting bias may be present. Many studies selectively reported only PPV or sensitivity, omitting specificity,

NPV, or confidence intervals, especially in settings where true negative denominators could not be reliably defined (e.g., in convenience samples or specialty cohorts). This selective reporting limits the interpretability of comparative performance across studies and inflates the apparent consistency of coding validity.

Finally, geographic clustering of studies in North America and Western Europe, with almost no representation from LMICs, indicates a broader structural bias in the availability and dissemination of EHR validation research. Although tools such as NLP, technology-enabled abstraction, and registry linkage are increasingly available, their deployment remains largely confined to well-resourced health systems, potentially skewing the evidence base toward high-income healthcare environments with stronger documentation infrastructure.

While there is no definitive statistical evidence of publication bias due to the non-quantitative nature of most validation studies, the structural features of the literature, including selective reporting, geographic concentration, and journal outlet characteristics, indicate a moderate risk of such bias influencing the field's collective understanding of diagnostic coding concordance.

Certainty of evidence

The certainty of evidence across the 17 included studies was evaluated using the GRADE framework, adapted for diagnostic test accuracy studies. Given the unique challenges of validating EHR-based diagnoses, the certainty of the findings reflects not only methodological rigor but also the consistency with which coding outcomes reflect clinical truth across diverse contexts. A Summary of Findings table, including GRADE ratings and rationale for downgrading decisions, is provided in Table 5.

Discussion

Principal findings

This systematic review and meta-analysis synthesised evidence from 17 validation studies evaluating the concordance between EHR-derived data, ICD-coded diagnoses, and evidence-based treatment guidelines across diverse clinical domains. Furthermore, the reliability of chart abstraction as a reference standard must be considered in light of documented biases in EHR systems. Three major themes emerge from the collective analysis

The first relates to the observed difference between sensitivity and specificity in ICD-coded diagnostic data. Across most studies, specificity and positive predictive value (PPV) were consistently high, often exceeding 90%, particularly in conditions with well-defined diagnostic criteria and repeated healthcare interactions, such as adult congenital heart disease, hepatocellular carcinoma, and cirrhosis. For instance, Wong et al. [19] reported a PPV of 97.4% for hepatocellular carcinoma using $\geq$10 ICD code entries, while Cohen et al. [26] found a PPV of 93% and specificity of 99% for congenital heart disease coding. In contrast, sensitivity varied considerably across conditions, with lower values observed in domains such as obesity, dyslipidaemia, and symptom-based presentations. This pattern should be interpreted in the context of the intended function of ICD coding systems. ICD classifications are designed to capture diagnoses that are clinically relevant to a specific encounter, particularly those influencing management, reporting, or reimbursement, rather than to provide a complete account of all existing patient conditions. As such, lower sensitivity does not necessarily indicate coding inaccuracy; rather, it reflects the bounded scope of classification systems relative to the broader, more inclusive nature of clinical documentation within EHRs.

The second major finding concerns the distortion of estimates of guideline-concordant treatment when relying solely on ICD-coded data. Multiple studies have demonstrated that diagnostic codes are frequently applied at the time of treatment initiation rather than during diagnostic evaluation, reflecting a "structural characteristic of coding systems": high adherence to clinical guidelines. For example, in bone metastases, 76.5% of patients received appropriate bone-targeting agents, as confirmed by technology-assisted chart abstraction, but ICD-based estimates falsely inflated this to 89.2%. Similarly, in obesity and soft-tissue sarcoma, coded treatment documentation overestimated adherence due to selective coding of treated patients. These discrepancies highlight the limitations of structured coding in capturing the full care continuum and underscore the importance of validating eligibility and treatment data against clinical records or registries.

A third core finding emerges from subgroup and sensitivity analyses, which revealed substantial variability in coding accuracy by patient demographics, healthcare settings, and algorithm design. Coding rates were disproportionately higher among older adults, women, and individuals with greater comorbidity. In obesity documentation, women were 34% more likely to receive ICD codes than men, while Native American and Hispanic patients were also more likely to be coded compared to White patients, independent of actual BMI. Furthermore, specialty care units such as cardiology and oncology consistently demonstrated higher diagnostic sensitivity than general or emergency care settings. Methodologically, algorithm performance improved markedly when using multi-code logic, NLP, or code-frequency thresholds, indicating that composite or hybrid phenotyping methods are more accurate than relying solely on single ICD codes.

Taken together, these findings demonstrate that ICD-coded administrative data are insufficient as standalone sources for epidemiologic surveillance, quality monitoring, or secondary research. The systematic underascertainment of disease and distortion of apparent guideline adherence have direct implications for artificial intelligence and machine learning applications that rely on EHR data for model training and validation. Algorithms trained on incompletely coded data risk inheriting and amplifying these documentation biases. Integrating structured codes with laboratory data, medication records, and narrative clinical text is therefore essential for improving both research validity and the safe deployment of data-driven clinical decision support tools.

Comparison with previous reviews

The findings of this review are consistent with, and in several cases extend, existing literature evaluating the diagnostic validity of EHRs and ICD-coded administrative data.

The systematic review and meta-analysis by Mwogosi and Simba [38] reported a 32% reduction in diagnostic errors and a 26% reduction in medication errors following EHR implementation, particularly in mature and standardized systems. These results support the overall utility of EHR systems in improving patient safety. However, the present review emphasizes that despite these benefits, structured ICD-coded data often underrepresents the true clinical burden of disease, especially in chronic and underdiagnosed conditions. This underscores a distinction between reductions in overt clinical error and limitations in data fidelity within administrative records.

Stanfill et al. [39] conducted an extensive systematic review of 113 studies on automated clinical coding systems. Their conclusions, highlighting a decline in system accuracy with increasing diagnostic complexity, are reflected in the observed variability in coding concordance across disease domains in this analysis. Several included studies demonstrated improved accuracy when combining structured codes with natural language processing (NLP), reaffirming the notion that ICD codes alone are insufficient for capturing multifaceted clinical presentations.

The study by Kamal et al. [40], which validated ICD-10 codes for hypoparathyroidism in a national patient registry, found a high positive predictive value (PPV) of 91%. Similar patterns of high PPV were noted in this review for conditions such as cirrhosis and hepatocellular carcinoma. However, the current synthesis also highlights substantial variation in sensitivity, with markedly low detection rates for obesity and dyslipidemia, indicating that high PPV does not necessarily equate to

comprehensive case identification.

Buford et al. [41] focused on EHR systems used to manage chronic diseases among refugee populations. Their thematic analysis emphasized the benefits of EHRs in facilitating screening and treatment linkage, while also highlighting the challenges posed by linguistic and infrastructural barriers. Although this review did not concentrate on displaced populations, parallels can be drawn regarding underdocumentation in vulnerable or underserved groups and the reliance on specialized care settings to ensure accurate coding.

This review corroborates previous evidence on the strengths of EHR systems in supporting clinical care and confirms known limitations in the diagnostic accuracy of structured coding. It further contributes by quantitatively synthesizing evidence on the performance of ICD codes across clinical domains, highlighting systematic underascertainment in certain conditions, and demonstrating the superior performance of hybrid phenotyping approaches involving NLP and algorithmic augmentation.

Study implications

The findings from this review reveal critical implications for healthcare practitioners, informaticians, and health system administrators involved in data-driven quality improvement and clinical decision-making. While ICD-coded administrative data offer a scalable and standardized means of tracking clinical activity, the observed limitations in sensitivity suggest that these codes frequently underreport the true burden of disease, particularly for chronic conditions, milder phenotypes, and underdiagnosed populations.

For clinicians, this has direct consequences in performance evaluation, risk stratification, and population health initiatives that rely on accurate diagnostic capture. The absence of a formal code can lead to missed opportunities for intervention, delayed treatment, or erroneous quality metrics that do not reflect the care actually provided. As demonstrated in several included studies, patients often receive treatment without corresponding diagnostic codes, whereas in other cases, diagnoses are coded only after treatment is underway, creating a misleading impression of high adherence.

Health informatics professionals should be cautious in using ICD codes as sole inputs for automated alert systems, registry identification, or research cohort construction. Given the consistent discrepancies between structured codes and clinical reference standards, structured data must be augmented with more nuanced approaches, including free-text analysis, procedure codes, medication data, and clinician notes. Institutions should also invest in clinician training and interface designs that support accurate and consistent coding practices, especially in generalist settings where documentation practices vary widely.

These findings emphasize the need to continue refining electronic phenotyping methods. Hybrid approaches that combine structured administrative codes with unstructured clinical data, particularly NLP techniques, show substantial promise for improving both sensitivity and precision in disease identification. Several high-performing studies in this review achieved strong predictive validity only after integrating NLP-derived concepts, frequency thresholds, or "either/or" coding algorithms.

Future research should prioritize validating these enhanced phenotyping methods across diverse populations and clinical settings. Importantly, very few studies have been conducted in LMICs, where infrastructure limitations, undercoding, and non-standardized EHR systems may compound documentation challenges. Expanding validation studies to these contexts is essential to ensure global generalizability of findings.

Another critical area involves the transition to ICD-11, which introduces greater granularity and post-coordination capabilities. While this may theoretically enhance diagnostic specificity, it also presents new challenges in code complexity, training, and interoperability. Evaluating the performance of ICD-11-based coding systems across a range of diseases and care settings should be a research priority as global adoption increases.

Study limitations

Despite the strengths of this review, including the use of validated reference standards and inclusion of diverse clinical domains, several limitations warrant consideration.

The evidence base remains disproportionately centered on high-income countries and academic health systems. As a result, the findings may not fully capture documentation patterns in community-based care or low-resource settings where EHR infrastructure, clinical coding practices, and workforce training can differ substantially.

The heterogeneity of study designs, target conditions, and outcome reporting posed a challenge for quantitative synthesis. Many studies did not report complete diagnostic accuracy metrics, such as specificity or negative predictive value, particularly those that employed case-enriched sampling. This limited the ability to perform pooled statistical analysis and may have introduced reporting bias.

Exclusion of unpublished studies, internal health system audits, or non-English literature with unavailable translations may have narrowed the scope of the review. Additionally, variation in reporting quality across included studies, such as missing inter-rater reliability or unclear blinding procedures, raises concerns about unmeasured internal bias, even when reference standards were robust.

The evolving landscape of diagnostic classification, including the introduction of ICD-11, also poses a challenge for long-term applicability. Many of the included studies assessed ICD-9 or ICD-10 coding, which may not reflect newer terminology or emerging digital abstraction tools now used in clinical settings.

An additional limitation concerns the use of chart review as the reference standard. Clinical documentation within EHRs is itself subject to well-recognised sources of bias, including copy-and-paste practices, propagation of outdated diagnoses, and inclusion of unverified clinical assertions. These factors may inflate the apparent prevalence of conditions in chart-reviewed data and introduce systematic bias into validation studies. As a result, discrepancies between ICD-coded data and reference standards may partially reflect imperfections in the reference standard itself, rather than true coding inaccuracies.

Conclusion

This systematic review highlights both the utility and limitations of structured ICD-coded data within EHRs as a proxy for clinical diagnoses and treatment documentation. While positive predictive values were generally high, suggesting reliability for "ruling in" disease, sensitivity was consistently low, particularly for chronic, undercoded, or symptom-based conditions. This "sensitivity-specificity gap" has significant implications for epidemiologic research, health services evaluation, and the generation of real-world evidence.

The findings underscore the need for methodological rigor when using EHR-derived administrative data. Reliance on diagnosis codes alone risks misclassification, underrepresentation of treated but uncoded patients, and inflated estimates of treatment adherence. Multimodal phenotyping approaches that combine structured codes with free-text data, laboratory results, and clinician narratives are essential to achieving more accurate and equitable representations of patient care.

These findings have important system-level implications. Health systems should incorporate routine coding audits, strengthen clinician training in diagnostic documentation, and adopt hybrid phenotyping methods when using EHR data for performance measurement or research. Policymakers should recognise that administrative data alone may misrepresent disease burden and treatment quality. Investments in validation infrastructure, documentation standards, and algorithm transparency are essential for ensuring that EHR data can reliably support clinical governance and population health analytics.

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Competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

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Augustino Mwogosi: Data curation, Formal analysis, Investigation, Writing – original draft. **Cesilia Mambile:** Data curation, Validation, Investigation, Writing – review & editing.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.hlpt.2026.101213.

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